Course Outline for CSE-202

Part A

1. Course Code: CSE-202

2. Course Title: Software Development with JAVA (Sessional)

3. Course Type: Core Course

4. Level/ Term: Level: 2 Term: I

5. Academic Session: 2020-21

6. Course Teacher: Md. Mynul Hasan, Assistant Professor, Dept. of CSE, CUET
Ashim Dey, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 1.5

9. Contact Hours: 3 hours of lab work per week

10. Total Marks: 150

11. Rational of the Course:

This course is designed to provide fundamental concepts of JAVA programming and the applications appropriate in real world. The course focuses on basic concepts of JAVA programming and techniques, and their cross-cutting relationship to multiple problem domains in software development. Students will use a standard IDE to reinforce concepts of java programming including AWT, java swing, inheritance etc to develop java applications. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Introduction to JAVA and JAVA Development Kit (JDK), Fundamental Programming with JAVA, Object Oriented Programming based project development with JAVA, Take appropriate steps to archive all the project goals.

12. Course Objectives:

(a) Familiarization with JAVA environment and JDK
(b) To able to implement any ideas using JAVA
(c) To learn in depth about the analysis and design procedure of a software
(d) To implement project in such a way that is suitable for end users.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Understand the objective, terminology and fundamental concepts of JAVA programing	C2
CLO2	Apply the features of JAVA programming to develop projects	C3
CLO3	Design and implement real life software using JAVA that is suitable for end users	C3

b) Mapping of CLO with PLO

No.	**(CLOs)**	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
1	CLO1	X											
2	CLO2		X										
3	CLO3			X									

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Course overview, Introduction to JAVA and JDK	• Lecture • Group Discussion	Lab Performance Assignment	CLO-1
Week - 02	Fundamental concepts of Java including AWT, Swing, GUI.	• Lecture • Demonstration • Group Discussion	Lab Performance Assignment	CLO-1
Week - 03	Project topic selection	• Lecture • Demonstration • Implementation	Assignment	CLO-2
Week - 04	Design and analysis of individual projects	• Lecture • Demonstration • Implementation	Lab Performance Lab Report	CLO-2
Week - 05	Evaluating project initialization	• Lecture • Demonstration • Implementation	Lab Performance Lab Report	CLO-2
Week - 06	Monitoring project progress	• Lecture • Demonstration • Group Discussion	Lab Performance Lab Report	CLO-3
Week - 07	File management system	• Lecture • Demonstration • Implementation	Lab Performance Lab Report	CLO-2

Week - 08	Basic data-structure	• Lecture • Group Discussion	Lab Performance Lab Report	CLO-2
Week - 09	Improving User interface	• Lecture • Demonstration • Implementation	Lab Performance Lab Report	CLO-2
Week - 10	Final Project Submission	• Group Discussion	Lab Performance Lab Report	CLO-3
Week - 11	Project Report Evaluation	• Assignments	Lab Performance Documentation	CLO-3
Week - 12	Presentation and evaluation of the project	• Group Discussion	Lab Performance Presentation Slide	CLO-3
Week - 13	Viva-voce & Quiz			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%

b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- JAVA: The Complete Reference, Ninth Edition by Herbert Schildt

2) Supplementary Readings

- Java Concurrency in Practice by Brian Goetz, Tim Peierls, Joshua Bloch, Joseph Bowbeer, David Holmes and Doug Lea.
- Learn Java GUI Applications; 11th edition: A JFC Swing Tutorial by Philip Conrod and Lou Tylee

3) Others

- Handout/lecture material provided by the course teacher